TESTING OF RESISTORS - SERIES AND PARALLEL

5

Aim:

To study the voltage and current level, when resistors are connected in series and parallel.

Apparatus / Components required

S.No	Name of the apparatus / Components	Range / value	Quantity
1.	Regulated power supply	0-30 V DC	1 No
2.	Voltmeter	0-30 V DC	3 No
3.	Ammeter	0-100 mA	4 No
4.	Resistors	Three different values	1 each

Resistors in series

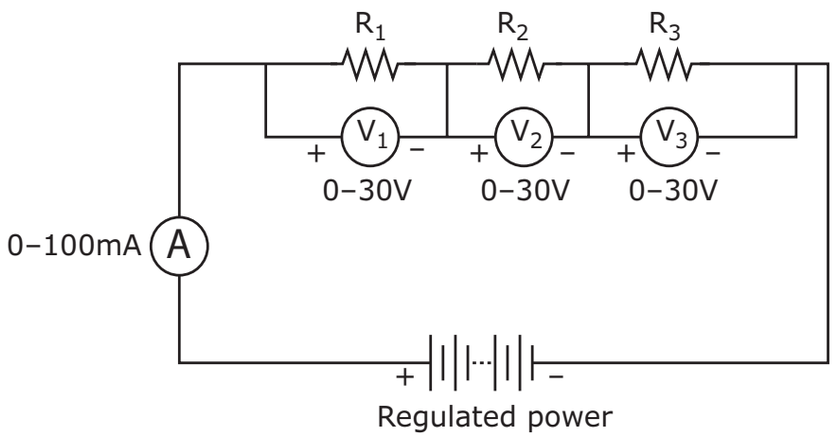


Fig 5,1 Resistors in Series

Given resistors	Value
R_1	
R_2	
R_3	
Total resistance	

Calculate the values

Procedure:

Resistors in series:

1. Connect the components as per the circuit diagram.
2. Switch ON the power supply.
3. Measure the voltage across each resistor (V_1,V_2,V_3).
4. Measure the total current (I).
5. Repeat the same procedure for different input DC voltages.
6. Tabulate the readings.

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